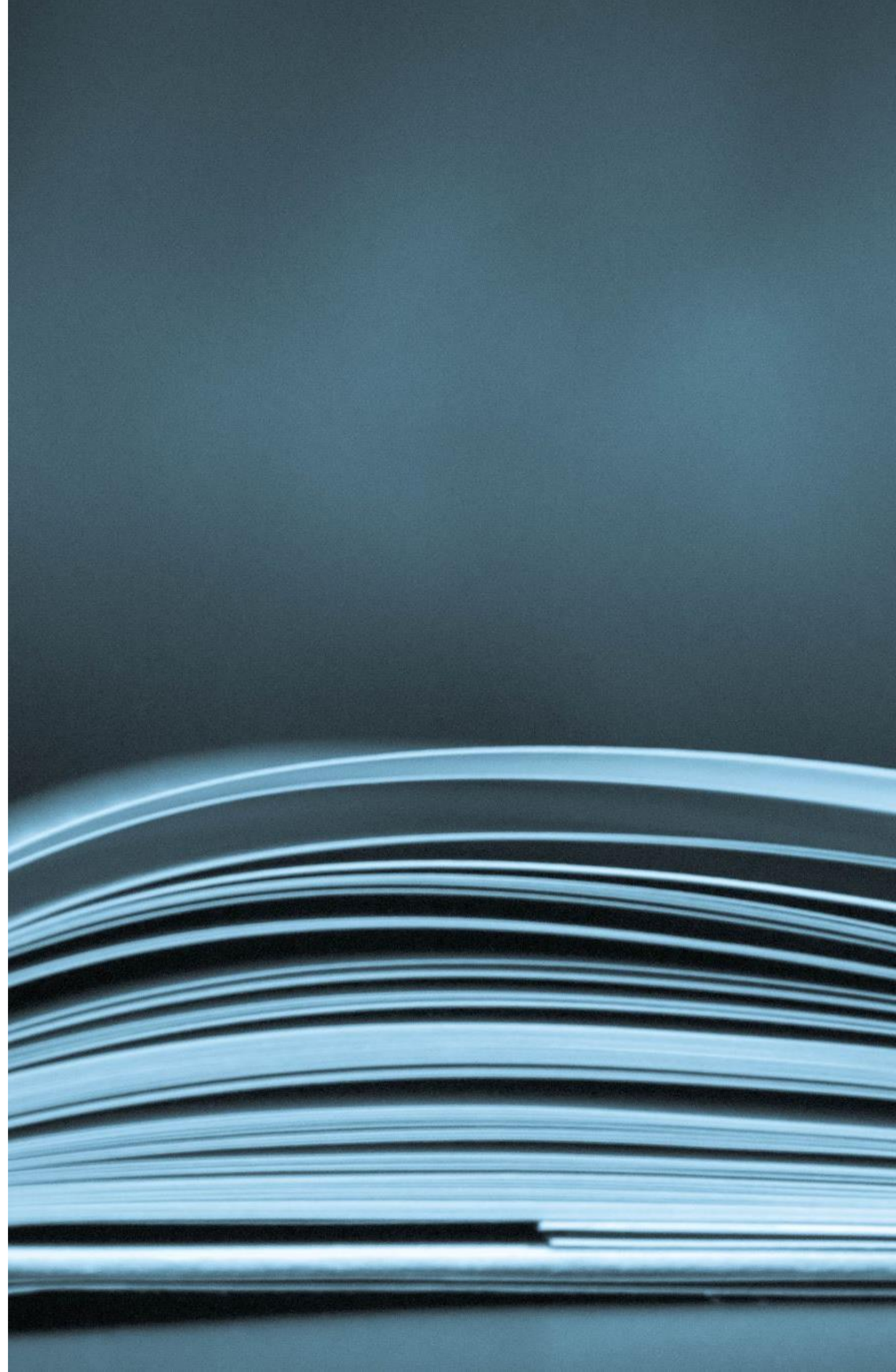
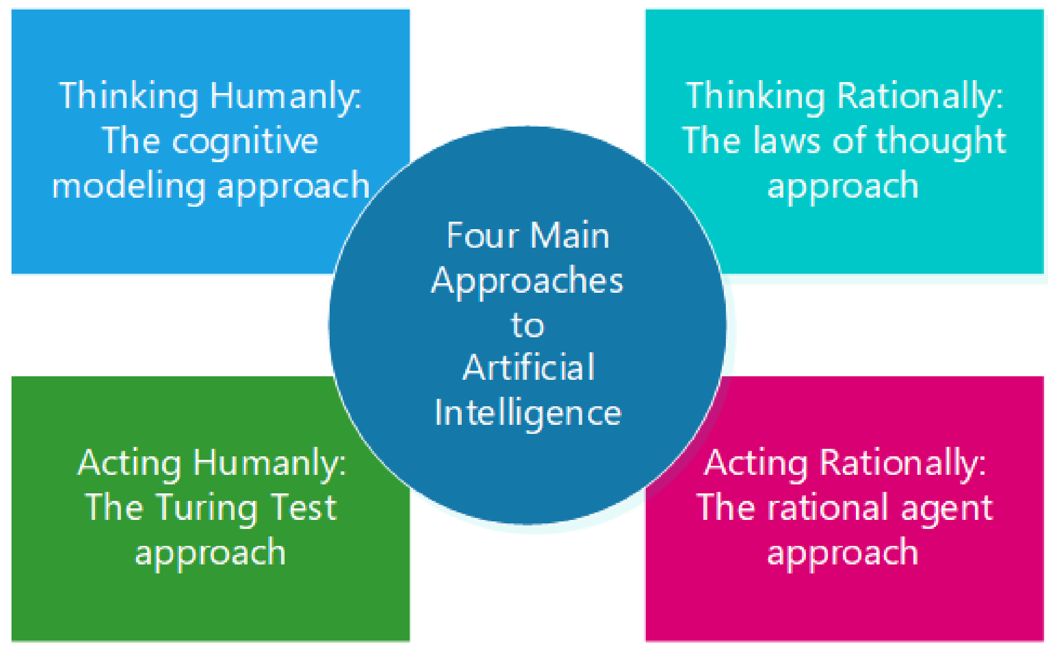
Course Title: Artificial Intelligence

- Course Code: CSC262 Course
- Title: Artificial Intelligence
- Credit Hours: 3 (2,1)
- Lecture Hours/Week: 2
- Lab Hours/Week:
- 3 Pre-Requisites: None
- Text and Reference Books
- Textbook: 1. Artificial Intelligence: A Modern Approach, Russell, S., and Norvig, P., Pearson, 2020. Reference Book: 1. Artificial Intelligence Basics: A Non-Technical Introduction, Taulli, T., Apress, 2019.



Four main approaches

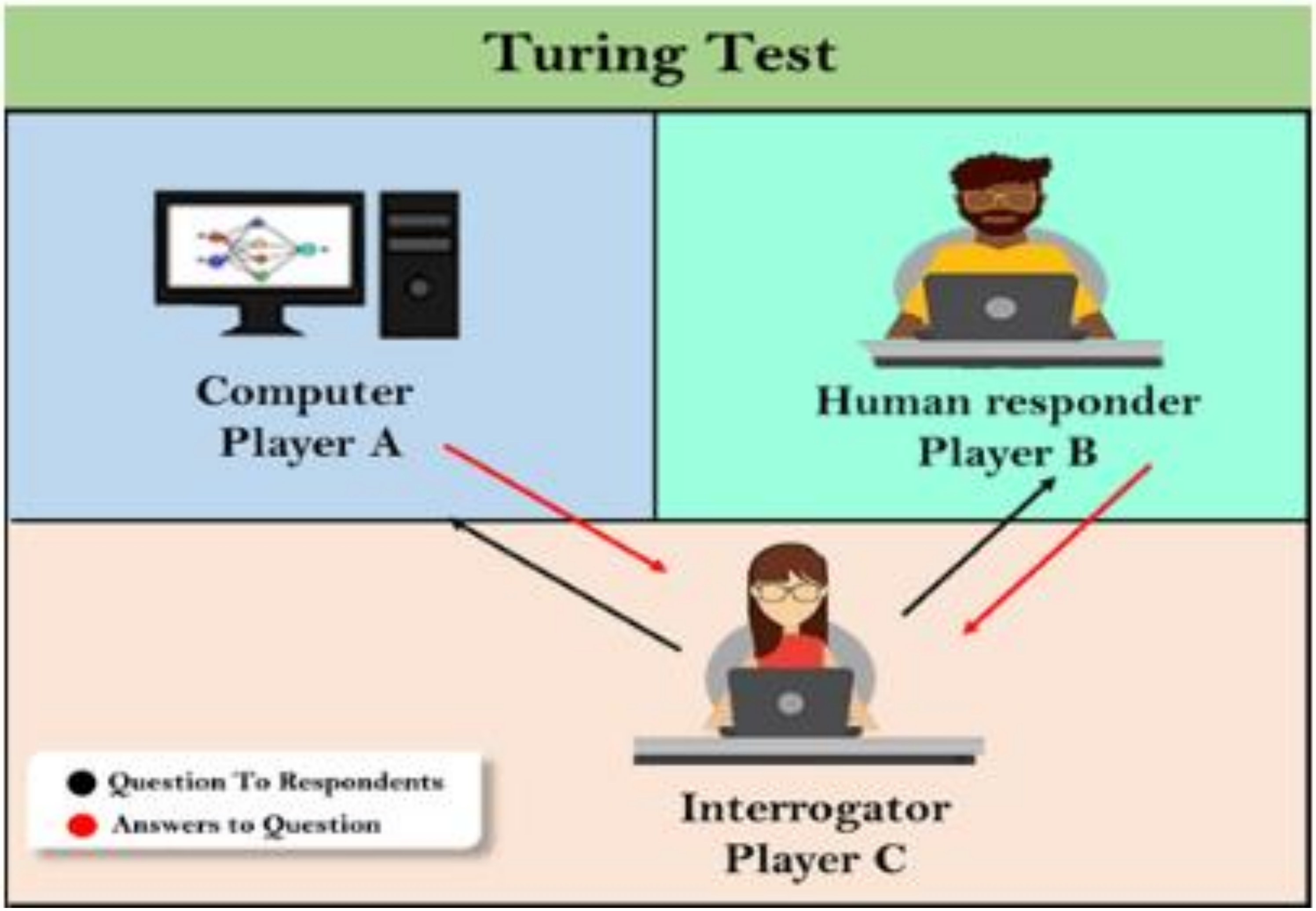


What is AI?

Acting humanly: **Acting like a human.**

- In 1950, **Alan Turing** introduced a test to check whether a **machine can think like a human or not**, this test is known as the ***Turing Test***. In this test, Turing proposed that the computer can be said to be an intelligent if it can mimic human response under specific conditions.
- Turning test is used to check whether a machine can think a human or not.
- The turning test require **three terminals**, each of which is **physically separated from the other two.**
- One terminal is operated by a **computer**, while the other **two are operated by human**.

.


Turing Test
Computer
Player A
Human responder
Player B
Interrogator
Player C
Question To Respondents
Answers to Question

Acting humanly

- **<u>Interrogator</u>** **Are you a Computer?**

- A (Computer) NO
- B (Human) NO

- In this game if an **interrogator** would not be able to identify which is a machine and which is human, then the computer **passes the test successfully, and the machine is said to be intelligent and can think like a human.**

Thinking Humanly: Cognitive Approach

- **Goal:** To create systems that think and reason like humans.
- **Focus:** Mimicking human thought processes, including learning, decision-making, problem-solving, and even emotions.
- **Methods:**
    - Cognitive modeling: Simulating human thought processes using computational models.
- **Example:** A **robot** that learns from its **environment**, makes decisions based on **past experiences**, and adapts its behavior like a human would.

Thinking Rationally: Laws of Thought

- **Goal:** To create systems that **use logical reasoning to solve problems.**
- **Focus:** Applying formal logic, rules, and structured reasoning to make decisions.
- **Methods:**
  - Theorem proving: Solving problems by applying logical rules step by step.
- **Example:** A **computer program that solves a math problem** by following **logical steps** or a **chess-playing AI that evaluates moves based on predefined rules.**

Acting Rationally: Rational Agent Approach

- **Goal:** To create systems that act in the best possible way to achieve their goals.
- **Focus:** Practical decision-making based on maximizing outcomes or utility, without necessarily mimicking human thought.
- **Methods:**
    - Rational agents: Systems that perceive their environment, take actions, and optimize outcomes based on goals.
    - Reinforcement learning: Learning through trial and error to maximize rewards.
- **Example:** A self-driving car that navigates traffic, avoids obstacles, and reaches its destination safely and efficiently.

Artificial Intelligence

INTELLIGENT AGENTS

Agents in Artificial Intelligence

Agent and Environment:

- An **agent** is anything that can perceive its environment through **sensors** and acts upon that environment through **effectors.**
- **An agent can be:**

1. Human-Agent

- eyes, ears, and other organs for sensors; – hands,legs, mouth, and other body parts for actuators

2.Robotic Agent

- cameras and infrared range finders for sensors; – various motors for actuators

3. Software Agent.

- Keystrokes, file contents, received network packages as sensors – Displays on the screen, files, sent network packets as actuators

Agents in Artificial Intelligence

- **Sensor:** Sensor is a **device** which detects the **change in the environment and sends** the information to other electronic devices**. An agent observes its environment through sensors.**
- **Actuators:** Actuators are the component of machines that converts energy into motion. **The actuators are only responsible for moving and controlling a system**. An actuator can be an electric motor
- **Effectors:** Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.

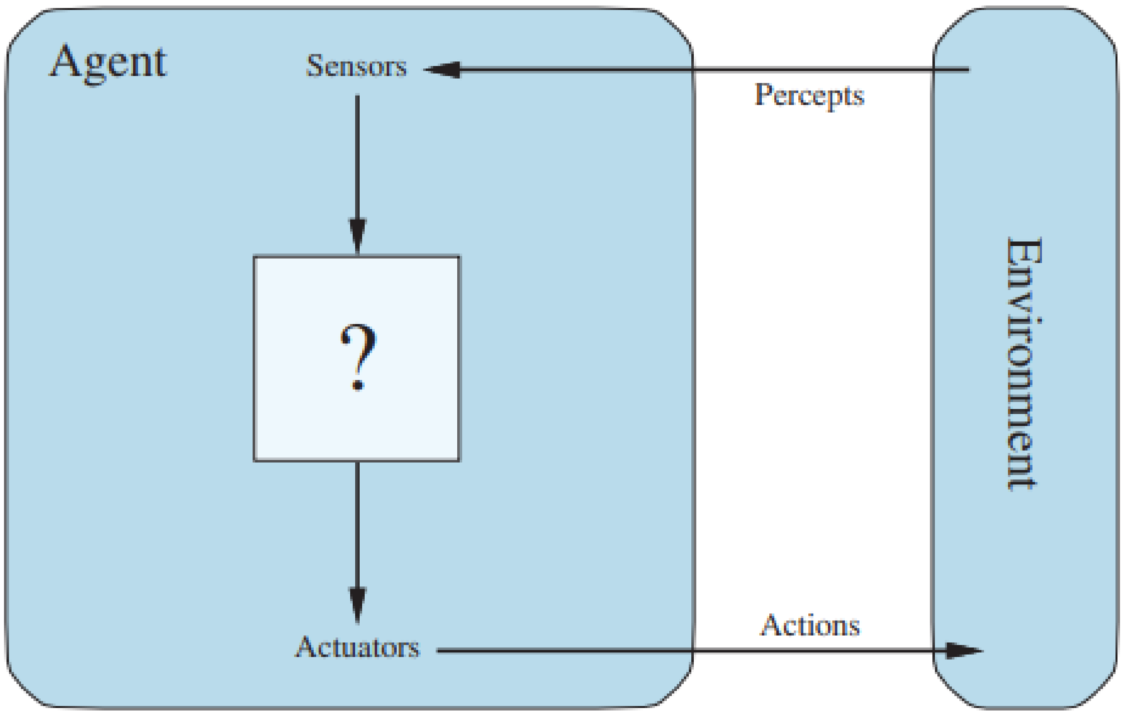


Figure 2.1 Agents interact with environments through sensors and actuators.

<u>Intelligent Agents:</u>

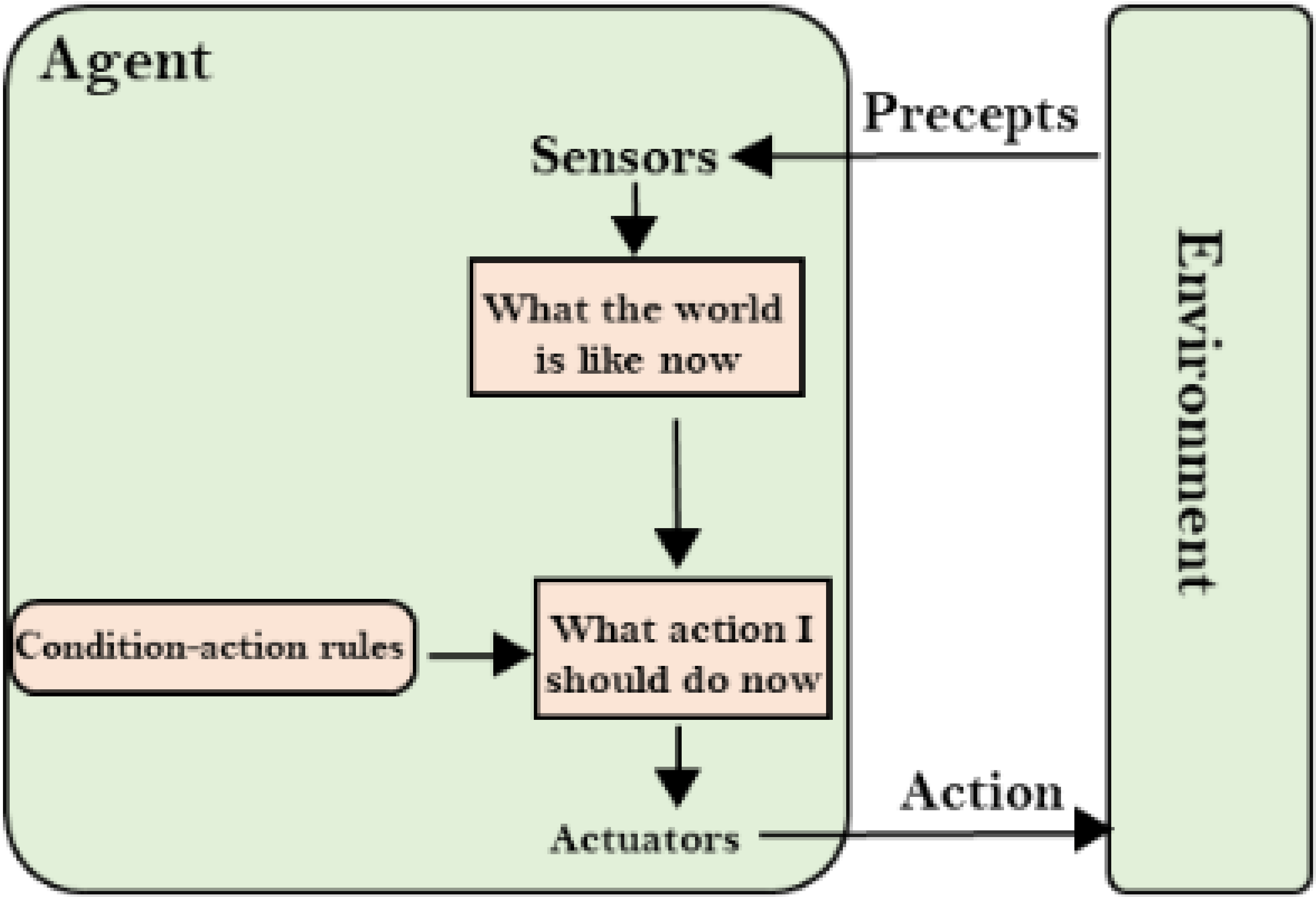
- An Intelligent agent is an **autonomous entity** which act upon an environment using **sensors** and **actuators** for achieving goals.
- An intelligent agent may learn from the environment to achieve their goals.

Agent Types

- Simple Reflex Agent
- Model-based reflex agent
- Goal-based agents
- Utility-based agent
- Learning agent

Simple Reflex Agents

- **How They Work:** These agents select actions based on the **current percept** (input from sensors) using **predefine rules (if-then rules).**
- **Strengths:** Simple and **fast for well-defined, predictable environments**.
- **Limitations:** Cannot handle environments with **partial observability** or require memory of past states.


Agent
Precepts
Sensors
What the world is like now
Condition-action rules
What action I should do now
Action
Actuators
Environment

Simple Reflex Agents

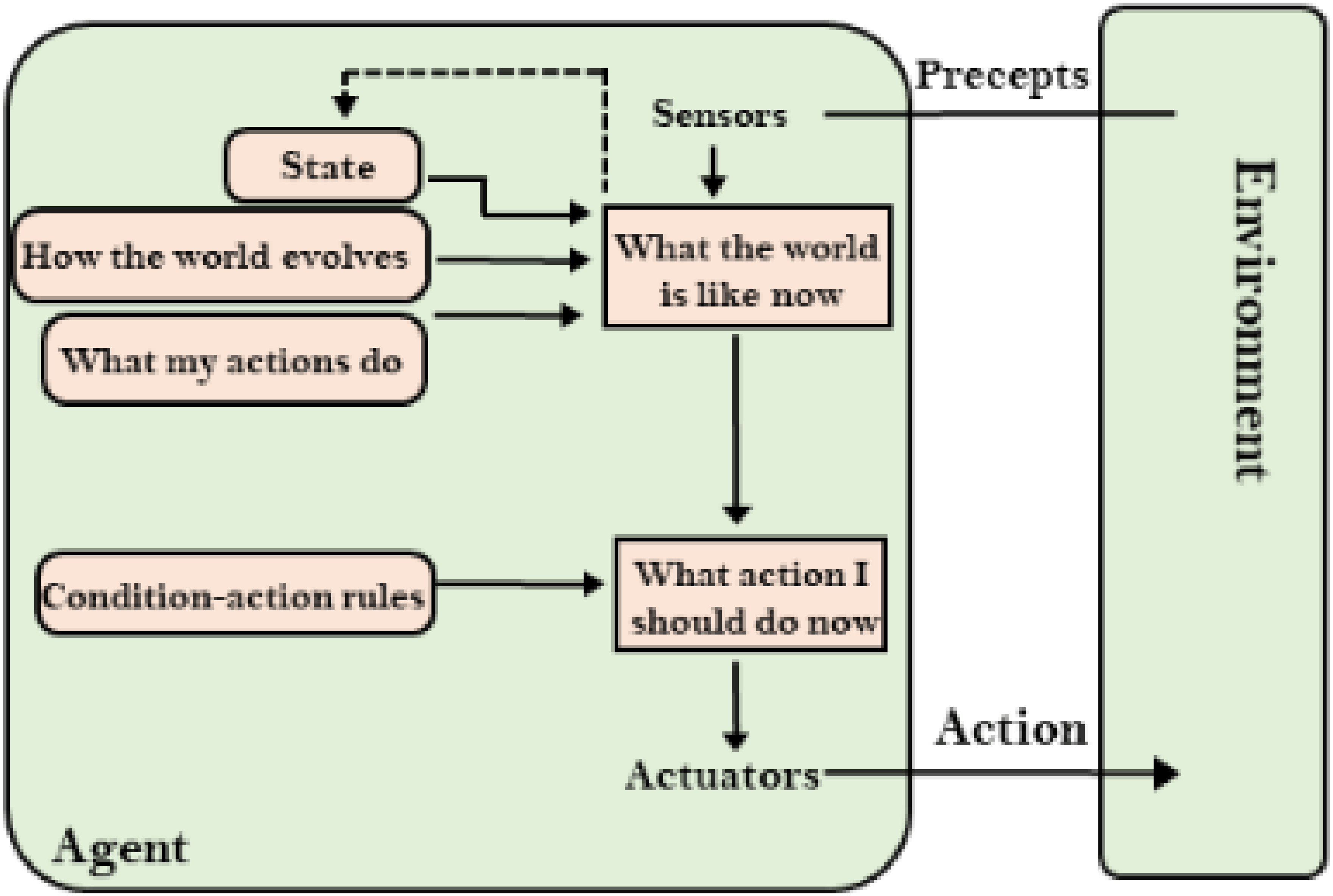
- **Light Sensor:** Turns on lights when it gets dark.
- **Automatic Door:** Opens when it detects motion.
- **Email Spam Filter:** Blocks emails containing specific keywords.
- **Smoke Detector:** Sounds an alarm when smoke is detected.
- **Traffic Light:** Changes signals based on predefined timers.
- **Automatic Fan:** Turns on when room temperature exceeds a limit.

Model-Based Reflex Agents

- **How They Work:** Reflex agent with state
- Have internal state which is used to keep track **of past states of the world**
- **Strengths:** Model-based agent can work in a **partially observable environment** and track the situation.
- **Limitations:** Still limited to reactive behavior (no long-term planning).

Model-Based Reflex Agents

- **Vacuum Cleaner:** Remembers which areas of a room have been cleaned.
- **Elevator Control System:** Tracks floor requests and optimizes movement.
- **Security System:** Activates alarms based on a sequence of sensor inputs.



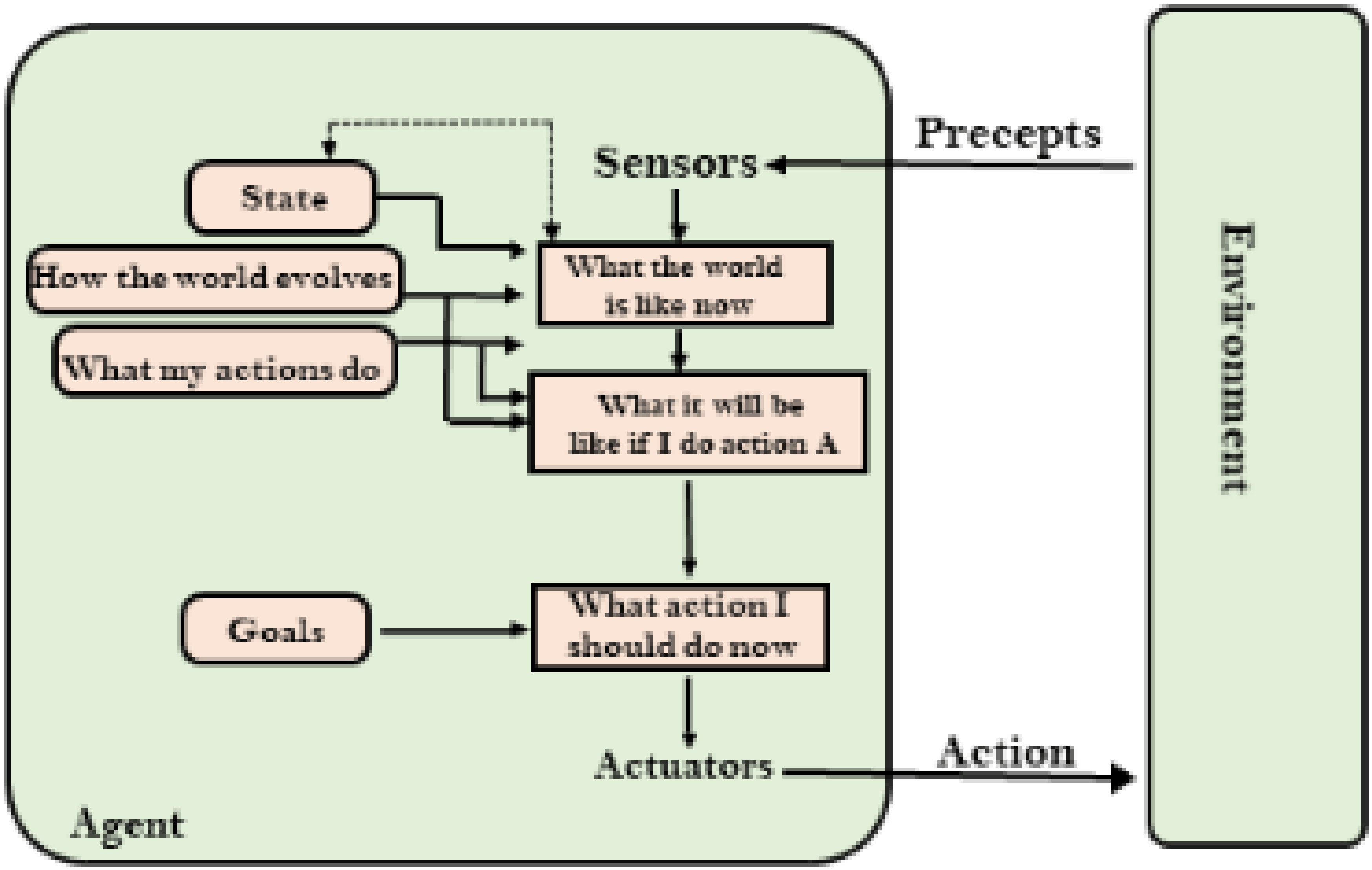
Precepts
Sensors
State
How the world evolves
What my actions do
What the world is like now
Condition-action rules
What action I should do now
Actuators
Action
Agent
Environment

Goal-Based Agents

- **How They Work:** These agents act to achieve specific goals. **They use search and planning algorithms to determine the sequence of actions needed to reach the goal.**
- **Strengths:** Can handle **complex tasks requiring planning** and decision-making.
- **Limitations:** May struggle in dynamic or uncertain environments.

Goal-Based Agents

- **Delivery Robot:** Plans the shortest route to deliver packages.
- **Route Planner:** Finds the fastest route for a self-driving car.
- **Warehouse Robot:** Picks and places items to fulfill orders.
- **Personal Scheduler:** Plans daily tasks and appointments.


Precepts
Sensors
State
How the world evolves
What the world
is like now
What my actions do
What it will be
like if I do action A
Goals
What action I
should do now
Actuators
Action
Agent
Environment

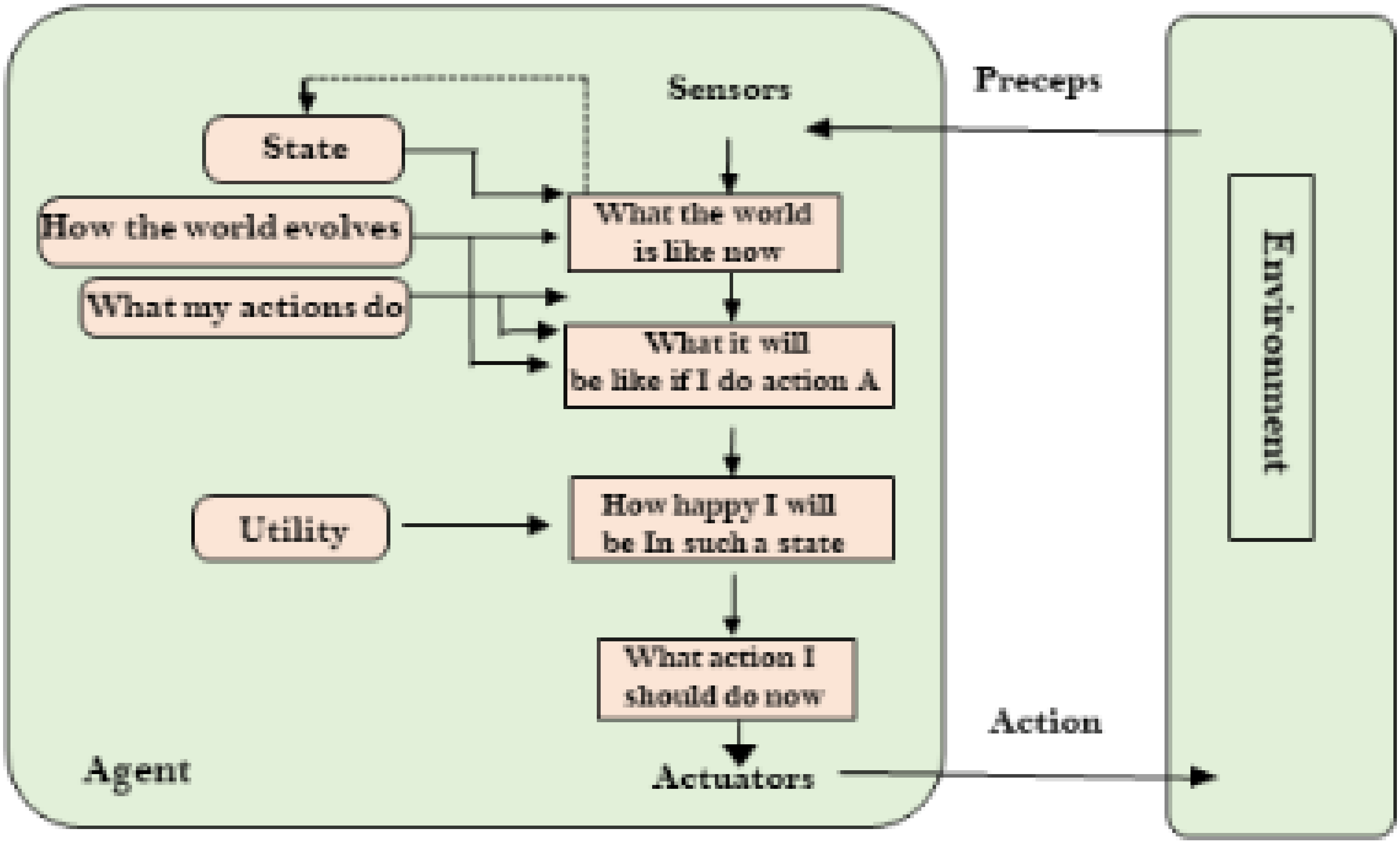
Utility-Based Agents

How They Work: These agents aim to maximize a utility function, which measures the **"happiness" or satisfaction** of the agent. They evaluate actions based on their expected **outcomes and choose the one with the highest utility**.

- **Strengths:** Can handle trade-offs and make decisions in complex, **uncertain environments.**
- **Limitations:** Designing an accurate utility function can be challenging.

Utility-Based Agents

- **Self-Driving Car:** Balances speed, safety, and fuel efficiency.
- **Stock Trading Bot:** Maximizes profit while minimizing risk.
- **Healthcare AI:** Recommends treatments based on patient outcomes.
- **Recommendation System:** Suggests products or content to maximize user satisfaction.
- **Smart Home System:** Adjusts lighting, temperature, and security to maximize comfort.

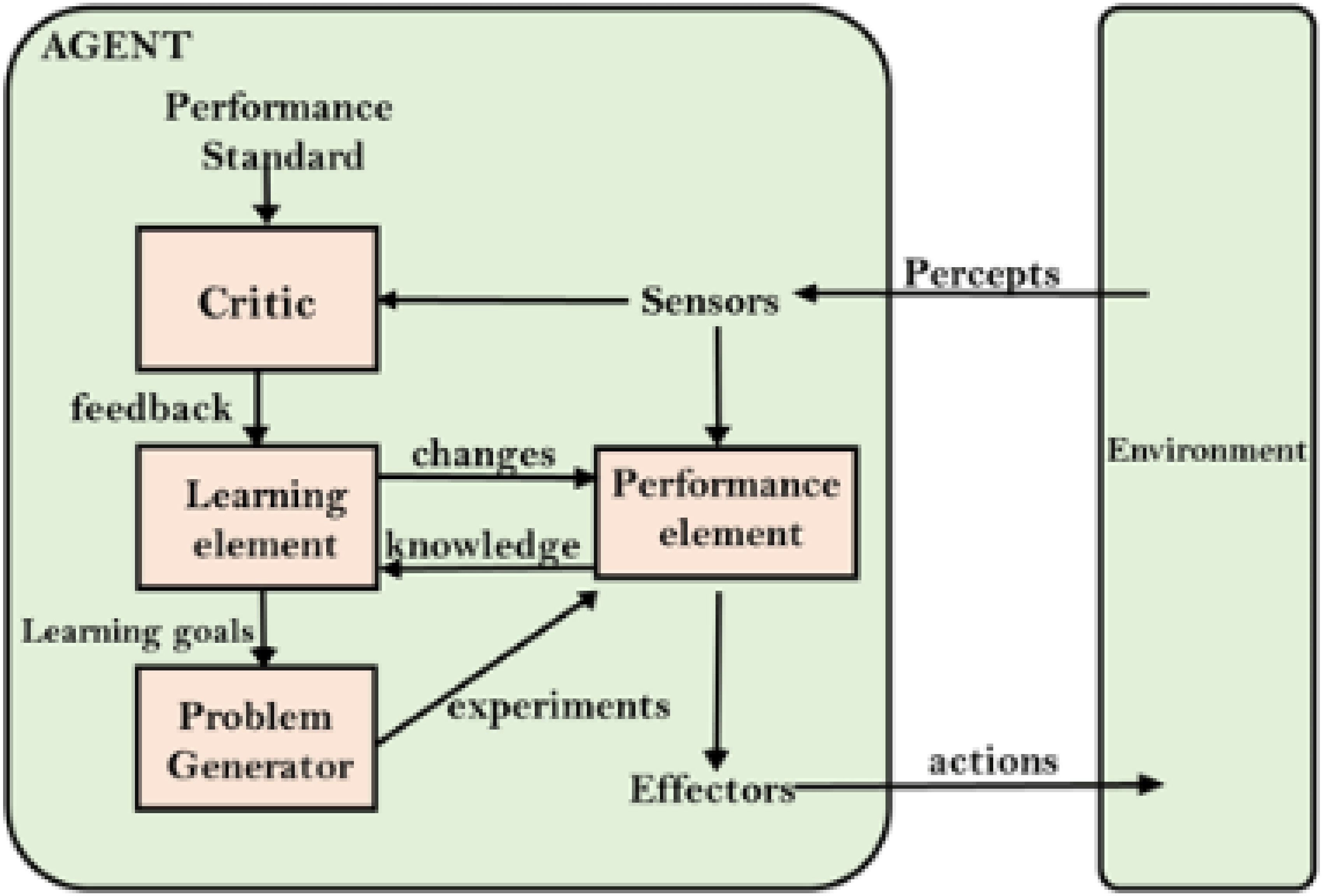

Sensors
Preceps
State
How the world evolves
What my actions do
What the world
is like now
What it will
be like if I do action A
Utility
How happy I will
be In such a state
What action I
should do now
Action
Agent
Actuators
Environment

Learning Agents

- **How They Work:** These agents improve their performance over time by learning from their experiences. **They consist of four components:**
  - **Learning Element:** Improves the agent's knowledge or behavior.
  - **Performance Element:** Selects actions based on the current knowledge.
  - **Critic:** Provides feedback on the agent's performance.
  - **Problem Generator:** Suggests new experiences for learning.
- **Strengths:** Adaptable and capable of improving in dynamic environments.
- **Limitations:** Requires time and data to learn effectively.

Learning Agents

- **Netflix Recommendation System:** Learns user preferences to suggest movies.
- **Spam Filter:** Improves accuracy by learning from flagged emails.
- **Fraud Detection System:** Identifies fraudulent transactions by learning patterns.
- **Autonomous Vehicle:** Improves driving by learning from real-world data.


AGENT
Performance Standard
Critic
Sensors
Percepts
feedback
changes
Learning element
Performance element
knowledge
Environment
Learning goals
Problem Generator
experiments
actions
Effectors